

Algorithmic Game: Assignment Report

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1. Problem Overview

The competition task is a **supervised regression problem**. Given social media engagement statistics for posts, follower count, likes, dislikes, effective love, shares, reposts, and comments, the model must predict a continuous **score** for each post in the test set.

The training set contains **7,000 samples** across 10 columns (including the target score). The test set contains **3,000 samples** without the target. Predictions must be submitted in the format defined by *samp_subm_3.csv* (columns: ID and score).

2. Dataset Description

Column	Type	Description
ID	int	Unique record identifier (not used as feature)
User_id	int	User identifier (not used as feature)
follower_count	int	Number of followers of the poster
likes	int	Total likes on the post
dislikes	int	Total dislikes on the post
effective_love	int	Net positive reactions (likes – dislikes)
shares	int	Number of times the post was shared
reposts	int	Number of reposts
comments	int	Total comments on the post
score	float	TARGET: continuous score to predict

No missing values were found in either the training or the test set, so no imputation was strictly necessary (a median imputer was included in the pipeline as a safety measure). All features are numeric integers.

Target Variable Statistics (train_3.csv)

Statistic	Value
Count	7,000

Mean	100.94
Std Dev	594.82
Min	-1,728.03
25th pct	-325.60
Median	95.86
75th pct	519.83
Max	1,856.19

3. Approach

Three scikit learn algorithms were benchmarked using a held out validation split (80/20) and 5 fold cross validation (metric: Root Mean Squared Error: RMSE). All models were evaluated on the same 7 engineered features.

Algorithm	Val RMSE	5 Fold CV RMSE	R ² Score
Ridge Regression	88.49	88.49	0.978
Random Forest Regressor	~41.00	95.00	0.995
Polynomial LR (degree=2)	~0.000	~0.000	1.000

Polynomial Linear Regression (degree=2) was selected as the final model because it achieved near zero error, revealing that the dataset was synthetically generated from a degree 2 polynomial formula.

4. Algorithm: Polynomial Linear Regression (Degree = 2)

A scikit learn **Pipeline** was used with the following four sequential steps:

Step	Component	Purpose
1	SimpleImputer (median)	Fill any missing values with the column median
2	StandardScaler	Normalise features to zero mean, unit variance
3	PolynomialFeatures (d=2)	Expand 7 features: 35 (squares + cross products)
4	LinearRegression	Fit a linear model on the expanded feature space

Why Polynomial Features?

With 7 original features, **PolynomialFeatures(degree=2)** generates 35 terms: the original 7 features, all 7 squared terms (x_i^2), and all 21 pairwise interaction terms ($x_i \times x_j$). LinearRegression then learns the exact coefficient for each of these 35 terms.

The dataset was synthetically created using a quadratic formula over the engagement features. By expanding the feature space to degree 2, the pipeline exactly reverse engineers that formula, hence the validation RMSE of effectively zero.

Pipeline Code (Python)

```
model = Pipeline([
    ('imputer', SimpleImputer(strategy='median')),
    ('scaler', StandardScaler()),
    ('poly', PolynomialFeatures(degree=2, include_bias=False)),
    ('regressor', LinearRegression())
])
```

5. Model Performance Metrics

The model was evaluated on a **20% held out validation set** (1,400 samples) and with **5 fold cross validation** on the full training set.

Final Result: Validation RMSE: **~0.000** | R² Score: **1.0000** | MAE: **~0.000**

Metric	Value	Interpretation
Validation MAE	~0.000	Mean absolute error: effectively zero
Validation RMSE	~0.000	Root mean squared error: effectively zero
Validation R ² Score	1.0000	Perfect fit: model explains 100% of variance
5 fold CV RMSE (mean)	~0.000	Consistent across all 5 folds: no overfitting
5 fold CV RMSE (std)	~0.000	Zero variance across folds
Training RMSE (full fit)	~0.000	Perfect in sample fit
Features (after poly expansion)	35	7 original: 35 polynomial terms

The RMSE values are not literally zero but on the order of 10^{-13} (floating point machine precision), which is consistent with the model having found the exact polynomial equation used to generate the data.

6. Prediction Screenshot (submission.csv: first 10 rows)

The table below shows the first 10 rows of the generated submission.csv, matching the required format (columns: ID and score).

ID	score (predicted)
7001	341.9618
7002	2.1468
7003	1317.8932
7004	-601.7638
7005	-491.7968

7006	670.2129
7007	23.7862
7008	-65.1530
7009	-195.2641
7010	991.5389

Full prediction stats (all 3,000 test rows): Mean = 97.81 | Std = 591.92 | Min = -1,672.03 | Max = 1,715.32

7. Conclusion

A scikit learn **Pipeline** combining StandardScaler, PolynomialFeatures(degree=2), and LinearRegression achieved a validation RMSE of effectively **zero** and an R^2 of **1.0**, outperforming both Ridge Regression and Gradient Boosting Regressor. The result demonstrates that the competition dataset was synthetically generated from a quadratic polynomial formula, which the model successfully identified and fitted exactly.